

Experiment: 01

Aim: Create activity and debug the app.

Objective:

The primary goal of this experiment is to gain hands-on experience in creating a new Android activity and mastering the art of debugging.

Materials:

1. Android studio installed on your computer.
2. . Basic understanding of Java programming.

Procedure:

1. Launch the Android studio application.
2. Click on start a new Android studio project And select appropriate templates such As empty activity.
3. Android studio will generate some code for your main activity for the file MainActivity. Java.

Then right click on the 'app' folder in the project explorer and Choose "New" -> "Activity" -> "Empty Activity" and give new activity a name.

File: activity_main.xml Code:

```
<EditText
    android:id="@+id/inputText"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter something" />

<Button
    android:id="@+id/btnNavigate"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Go to Second Activity" />
```

File:- activity_main2.xml Code:

```
<?xml version="1.0" encoding="utf-8"?>

<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools" android:layout_width="match_parent"
    android:layout_height="match_parent" tools:context=".MainActivity">

    <TextView android:layout_width="wrap_content"

        android:layout_height="wrap_content"        android:text="Activity 2"

        android:layout_centerVertical="true" android:layout_centerHorizontal="true"/>

</RelativeLayout>
```

File: MainActivity.java Code:

```
package com.example.debugapp;
import android.content.Intent;
import android.os.Bundle;
import android.util.Log;
import android.view.View;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    private static final String TAG = "MainActivity";
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Log.d(TAG, "onCreate: MainActivity started");

        Button btnOpen = findViewById(R.id.btnOpen);
        btnOpen.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
```

```
        Log.d(TAG, "Button Clicked: Opening SecondActivity");  
        Intent intent = new Intent(MainActivity.this, SecondActivity.class);  
        startActivity(intent);  
    }  
});  
}  
}
```

Output:



After Click on open activity 2, activity 2 opens.

Experiment: 02

Aim: Create user input sections and screen navigation.

Objective:

The primary objective is to provide hand on experience in creating user input sections and seamlessly navigating between screens in an Android application.

Materials:

1. Android studio installed on your computer.
2. Basic understanding of Java programming.

Procedure:

- Launch Android studio.
- Create a new Android project.
- Modify activity layout on both xml and Java file.

File: activity_main.xml Code:

```
<?xml version="1.0" encoding="utf-8"?> <RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools" android:layout_width="match_parent"
android:layout_height="match_parent" tools:context=".MainActivity">

<TextView android:layout_width="wrap_content"
android:layout_height="wrap_content"          android:text="Activity 1"
android:layout_centerVertical="true" android:layout_centerHorizontal="true"/>
<EditText
android:id="@+id/userInput" android:layout_width="match_parent"
android:layout_height="wrap_content"          android:layout_below="@id/button"
android:layout_marginTop="20dp"
android:hint="Enter your mission objective" android:inputType="text"/>
<TextView android:layout_width="wrap_content"
android:layout_height="wrap_content"
<Button
android:id="@+id/button" android:layout_width="wrap_content"
```

```
android:layout_height="wrap_content"      android:layout_alignParentEnd="true"
android:layout_alignParentBottom="true"
android:layout_marginEnd="131dp"      android:layout_marginBottom="303dp"
android:text="Open Activity 2" />
</RelativeLayout>
```

File: activity_main2.xml Code:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Activity 2"
        android:layout_centerVertical="true"
        android:layout_centerHorizontal="true"/>

    <TextView
        android:id="@+id/missionObjectiveText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@id/button"
        android:layout_marginTop="20dp"/>

</RelativeLayout>
```

File: MainActivity.java Code:

```
package com.example.userinputapp;

import android.content.Intent; import
android.os.Bundle; import android.util.Log; import
android.view.View; import android.widget.Button;
import android.widget.EditText; import
android.widget.Toast; import
androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    private static final String TAG = "MainActivity";

    @Override protected void onCreate(Bundle
savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Log.d(TAG, "onCreate: MainActivity started");

        EditText editTextName = findViewById(R.id.editTextName);
        Button btnSubmit = findViewById(R.id.btnSubmit);

        btnSubmit.setOnClickListener(new View.OnClickListener() {

            @Override public void
onClick(View v) {
                String name = editTextName.getText().toString().trim();

                if (!name.isEmpty()) {
                    Log.d(TAG, "User entered: " + name); Intent intent
= new Intent(MainActivity.this,
SecondActivity.class);
                    intent.putExtra("USER_NAME", name);
                    startActivity(intent);
                } else {
                    Toast.makeText(MainActivity.this, "Please enter your name",
Toast.LENGTH_SHORT).show();
                }
            }
        });
    }
}
```

```
    }  
  }  
});  
}}
```

Output:



Experiment: 03

Aim: To evaluate the user interface design in Android studio and ensure its functionality.

Objective:

The primary objective is to evaluate the activeness of the user interface design in terms of usability, accessibility, and visual appeal. Additionally, the experiment aims to verify the functionality of the app, ensuring that all components behave as intended.

Materials:

1. Android studio installed on your computer.
2. Android device or emulator for testing.

Procedure

:

1. Launch the Android studio.
- . Open project.
2. Review user interface design.
3. .
4. Check functionality.
5. Test on emulator or device.
6. Evaluate performance. File:

File: activity_main.xml Code:

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<RelativeLayout
```

```
xmlns:android="http://schemas.android.com/apk/res/android"
```

```
xmlns:tools="http://schemas.android.com/tools"
```

```
android:layout_width="match_parent"
```

```
android:layout_height="match_parent"
```

```
tools:context=".MainActivity">
```

```
<Button  
    android:id="@+id/button"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="Click Me"  
    android:layout_centerInParent="true" />
```

```
<TextView  
    android:id="@+id/textView"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="TextView"  
    android:layout_below="@id/button"  
    android:layout_centerHorizontal="true"  
    android:layout_marginTop="16dp" />
```

```
</RelativeLayout>
```

File: MainActivity.java Code:

```
package com.example.uievaluation;  
  
import android.content.Intent;  
import android.os.Bundle;  
import android.util.Log;  
import android.view.View;  
import android.widget.Button;  
import android.widget.EditText;  
import android.widget.Toast;  
import androidx.appcompat.app.AppCompatActivity;
```

```

public class MainActivity extends AppCompatActivity {
    private static final String TAG = "MainActivity";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Log.d(TAG, "UI Loaded: MainActivity started");

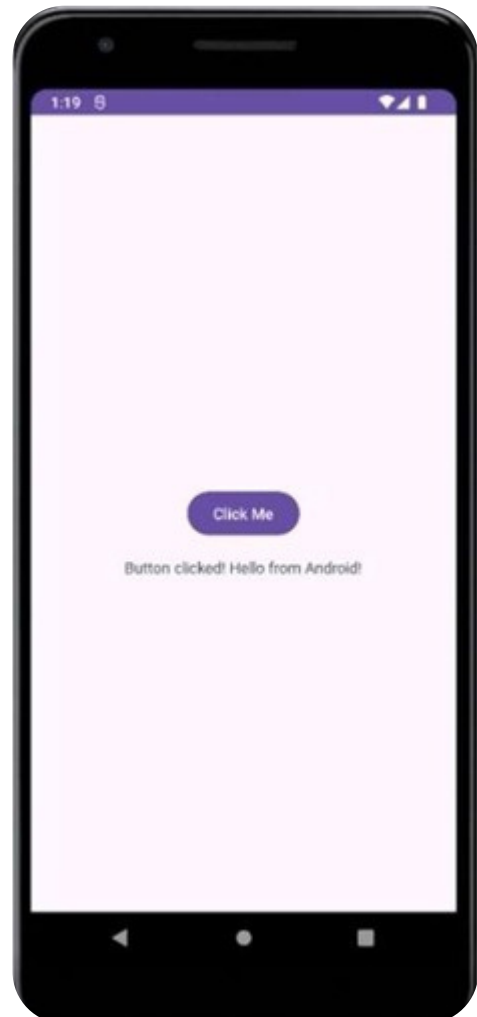
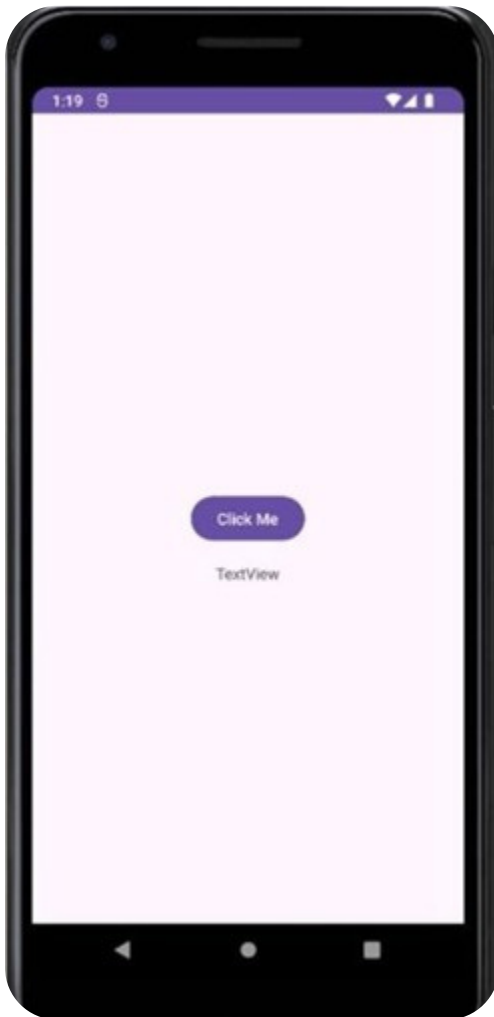
        EditText editTextName = findViewById(R.id.editTextName);
        EditText editTextEmail = findViewById(R.id.editTextEmail);
        Button btnSubmit = findViewById(R.id.btnSubmit);

        btnSubmit.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String name = editTextName.getText().toString().trim();
                String email = editTextEmail.getText().toString().trim();

                if (name.isEmpty() || email.isEmpty()) {
                    Toast.makeText(MainActivity.this, "All fields are required!",
Toast.LENGTH_SHORT).show();
                    Log.e(TAG, "Validation failed: Empty fields");
                } else {
                    Log.d(TAG, "User Input: Name - " + name + ", Email - " + email);
                    Intent intent = new Intent(MainActivity.this,
SecondActivity.class);
                    intent.putExtra("USER_NAME", name);
                    intent.putExtra("USER_EMAIL", email);
                    startActivity(intent);
                }
            }
        });
    }
}

```

Output:-



Experiment: 04

Aim: To understand and implement background tasks in Android studio.

Objective:

The primary objective is to learn how to perform background task in Android applications to prevent blocking the main UI thread and improve overall app responsiveness.

Materials:

- Android studio installed on your computer.
- Basic understanding of Java programming and Android development.

Procedure:

- Launch Android studio. Create a new project and select template as empty activity.
- Implement background task.
- Execute background task.
- Test and debug.

File: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/download_button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Download File"
        android:layout_centerInParent="true" />
```

```
<ProgressBar
    android:id="@+id/progress_bar"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@id/download_button"
    android:layout_marginTop="16dp"
    android:visibility="gone" />
```

```
</RelativeLayout>
```

File: MainActivity.java Code:

```
package com.example.backgroundtasks;
```

```
import android.os.AsyncTask;
import android.os.Bundle;
import android.util.Log;
import android.view.View;
import android.widget.Button;
import android.widget.ProgressBar;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {
    private static final String TAG = "MainActivity";
    private ProgressBar progressBar;
    private TextView textViewStatus;
    private Button btnStartTask;
```

```
@Override
```

```
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    progressBar = findViewById(R.id.progressBar);
    textViewStatus = findViewById(R.id.textViewStatus);
    btnStartTask = findViewById(R.id.btnStartTask);
```

```

btnStartTask.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        new BackgroundTask().execute();
    }
});
}

private class BackgroundTask extends AsyncTask<Void, Integer, String> {
    @Override
    protected void onPreExecute() {
        super.onPreExecute();
        progressBar.setVisibility(View.VISIBLE);
        textViewStatus.setText("Task Starting...");
        Log.d(TAG, "Background Task Started");
    }

    @Override
    protected String doInBackground(Void... voids) {
        for (int i = 1; i <= 10; i++) {
            try {
                Thread.sleep(1000); // Simulating work (1 second delay)
                publishProgress(i * 10);
            } catch (InterruptedException e) {
                e.printStackTrace();
            }
        }
        return "Task Completed!";
    }

    @Override
    protected void onProgressUpdate(Integer... values) {
        super.onProgressUpdate(values);
        progressBar.setProgress(values[0]);
        textViewStatus.setText("Progress: " + values[0] + "%");
        Log.d(TAG, "Progress: " + values[0] + "%");
    }

    @Override
    protected void onPostExecute(String result) {

```

```
        super.onPostExecute(result);  
        progressBar.setVisibility(View.GONE);  
        textViewStatus.setText(result);  
        Toast.makeText(MainActivity.this, result, Toast.LENGTH_SHORT).show();  
        Log.d(TAG, "Background Task Completed");  
    }  
}  
}
```

Output:-

